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Hi Fahad,

I want to give you a clear and detailed summary of the current progress on the AI marketplace project so you understand exactly where things stand.

First, the technical infrastructure for the system is now successfully set up and working on our server. We have deployed the entire platform and confirmed that the main components are communicating with each other correctly.

The system currently runs on our dedicated server with the following resources: 10 CPU cores, 28 GB RAM, and 320 GB SSD storage. This is powerful enough to run our own AI model locally without depending on external AI services.

The system architecture now includes several key components working together:

1. AI Language Model

We installed and are running an AI model called Qwen 2.5 on our server using Ollama. This model is responsible for understanding user messages and generating conversational responses. The AI can communicate in Arabic and English and will act as the “assistant” inside the marketplace.

2. Chat System

We built a chat interface where users can type requests such as:

“Show me phones”

“I want a used iPhone”

“Do you have cheap laptops?”

The message is sent to our backend server and processed by the AI model.

3. Product Database

All marketplace products will be stored in a database (PostgreSQL). This database holds the product details such as title, description, price, category, and seller information.

4. Smart Product Search (Vector Search)

We installed a system called Qdrant. This is a vector search engine that allows the AI to search products based on meaning, not just keywords.

For example:

If someone writes:

“I want a cheap Apple phone”

The system can still find products like:

“Used iPhone 11 – good condition”

Even if the words are slightly different.

5. AI Embeddings System

To make smart search possible, we use an embeddings model called nomic-embed-text. This converts product descriptions into mathematical vectors that allow semantic searching.

6. Streaming AI Responses

We confirmed that the AI responses stream live back to the user. This means the answer appears word-by-word like ChatGPT instead of waiting for a full response. This confirms the AI pipeline is working correctly.

We performed multiple tests on the server and confirmed that:

- The AI model is responding correctly.
- The chat system is working.
- The server is able to communicate with the AI model.
- The system is able to stream responses live.
- The backend API is working correctly.

This means the core AI infrastructure is now operational.

One important design decision we made is that the AI runs entirely on our own server instead of using paid external APIs like OpenAI or Google. This has several advantages:

- Lower operational cost in the long term
- Full control over the AI system
- No external API limits
- Better privacy for user data
- Ability to customize the AI for the marketplace

At the moment, the chat system is able to respond normally to messages such as greetings or general questions.

The next phase of development is to connect the AI directly with the marketplace product data so that it can act as a shopping assistant.

For example, when a user writes something like:

“I want a used iPhone under 200 KD”

The AI will:

1. Understand the request
2. Search the product database
3. Filter products by price and category
4. Return matching listings directly in the chat

This will allow users to interact with the marketplace entirely through conversation instead of traditional search filters.

In summary, the technical foundation of the AI marketplace is now fully running on our server. The AI chat engine, database, vector search system, and server infrastructure are all deployed and communicating correctly.

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The remaining work is mainly focused on improving the AI's ability to understand shopping requests and connect them with the actual product listings in the marketplace.

Once that is completed, users will be able to buy and sell products using a conversational AI assistant.

I will keep you updated as we move into the next stage.



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